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Building Knowledge Graphs

A Practitioner's Guide

Compliments of

neo4j

Jesús Barrasa
& Jim Webber

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Building Knowledge Graphs

Incredibly useful, knowledge graphs help organizations keep track of medical research, cybersecurity threat intelligence, GDPR compliance, web user engagement, and much more. They do so by storing interlinked descriptions of entities—objects, events, situations, or abstract concepts—and encoding the underlying information. How do you create a knowledge graph? And how do you move it from theory into production?

Using hands-on examples, this practical book shows data scientists and data engineers how to build their own knowledge graphs. Authors Jesús Barrasa and Jim Webber from Neo4j illustrate common patterns for building knowledge graphs that solve many of today's pressing knowledge management problems. You'll quickly discover how these graphs become increasingly useful as you add data and augment them with algorithms and machine learning.

- **Learn knowledge graph organizing principles**
- **Explore graph databases and knowledge graphs**
- **Import structured and unstructured data**
- **Build integration-and-search knowledge graphs**
- **Discover pattern detection knowledge graphs**
- **Use natural language knowledge graphs and chatbots**

Dr. Jesús Barrasa, an expert in semantic technologies and graph databases, is head of the solutions architecture team in EMEA at Neo4j and leads the development of neosemantics (a Neo4j plugin for RDF). He cowrote *Knowledge Graphs* (O'Reilly) and is cohost of the *Going Meta* live webcast.

Dr. Jim Webber is chief scientist at Neo4j, where he works on fault-tolerant graph databases. He coauthored *Graph Databases for Dummies* (Wiley) and *Graph Databases* and *Knowledge Graphs*, both for O'Reilly. He's also a visiting professor at Newcastle University, UK.

"An expert guide for engineers building or starting to work with knowledge graphs!"

—**Luanne Misquitta**
CTO, GraphAware

"This must-have book is the missing manual for working with graph data. It's a fun and incredibly useful guide. I highly recommend it."

—**Pete Aven**
Director/Solution Architect, MariaDB

"This book is outstanding for both beginners embarking on their knowledge graph journey and experienced graph enthusiasts seeking to elevate their skills and understanding."

—**Tareq Abedrabbo**
Data CTO, Mesh AI

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ISBN: 978-1-098-12711-4

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Beijing • Boston • Farnham • Sebastopol • Tokyo

Building Knowledge Graphs

by Jesús Barrasa and Jim Webber

Printed in the United States of America.

Published by O'Reilly Media, Inc., 1005 Gravenstein Highway North, Sebastopol, CA 95472.

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June 2023: First Edition

Revision History for the First Edition

2023-06-21: First Release

978-1-098-12711-4

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Preface

Graph databases and graph data science have reached a significant level of adoption. They have been extensively used for a range of discrete use cases like logistics, recommendations, and fraud detection. But there is a bigger emerging trend to arrange data in a deliberate manner that enables insight at scale across functional silos. The technology underpinning this trend is know as a knowledge graph.

The forces behind the trend are clear: organizations are no longer suffering from data scarcity. In fact, in an era when big data seems to be a solved problem (at least from a storage point of view), many organizations are practically drowning in data. Industry anecdotes of many thousands of relational tables per day being ingested into a data lake abound, but with an abundance of data there comes the unexpected challenge of what to with it. This is where knowledge graphs help.

A *knowledge graph* is a purposeful arrangement of data such that information is put in context and insight is readily available. Individual records are placed in an associative network of relationships that provide rich semantic connectivity and context. That network of relationships—a graph—is an incredibly intuitive way of representing useful knowledge. Data that might have originally existed to serve a fraud-detection use case can be repurposed seamlessly within the knowledge graph to provide data for recommending financial products (or vice versa). And from there it is straightforward to connect other data to support other vertical use cases or horizontal analysis.

Importantly, while the term *knowledge graph* has only come to prominence in industry relatively recently, knowledge graph systems have been in existence for some time. This book tries to distill our experience of understanding knowledge graphs deployed in real systems by organizations around the world. It addresses the emerging trend of building systems on knowledge graphs as well as thinking about knowledge graphs as a general-purpose underlay for the enterprise. It also addresses the contemporary intersection of knowledge graphs and artificial intelligence (AI), where knowledge graphs provide high-quality features for machine learning, are themselves enriched by AI, and can even tame the hallucinatory nature of large language models (LLMs).

While this is our most in-depth and unapologetically technical book on the topic, this isn't our first time writing about knowledge graphs. In fact, in the book *Knowledge Graphs: Data in Context for Responsive Businesses* (O'Reilly), we highlighted the business benefits of knowledge graph adoption aimed at an audience of CIOs and CDOs. But this book goes much deeper technically, and it contains enough implementation detail for a range of tools, patterns, and practices so that you can build your own knowledge graphs with confidence. We hope that what you learn here will propel you to your first successful knowledge graph project and beyond!

Who This Book Is For

This is a technical book, aimed at computing professionals—typically software engineers, system architects, and techincal managers—who want to understand both the potential of knowledge graphs and how to go about implementing them. While no prior experience with knowledge graphs (or graphs in the general sense) is required, readers will get the most of the book if they are modestly comfortable with database concepts like queries and have some programming experience.

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic
: Indicates new terms, URLs, email addresses, filenames, and file extensions.

`Constant width`
: Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

`Constant width bold`
: Shows commands or other text that should be typed literally by the user.

`Constant width italic`
: Shows text that should be replaced with user-supplied values or by values determined by context.

This element signifies a tip or suggestion.

This element signifies a general note.

This element indicates a warning or caution.

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Acknowledgments

We are grateful to all those who helped us while we were writing this book: the staff at O'Reilly, especially Corbin Collins, whose efforts kept our prose properly Americanized, and our Neo4j colleagues, especially Maya Natarajan and Deb Cameron, who worked with us on early drafts.

We are grateful to Dr. Nicola Vitucci, who provided deep technical expertise and patient guidance on our prose.

We would also like to thank Nigel Small, who provided detailed technical feedback on idiomatic Python graph data science.

Finally, we would like to thank the technical reviewers from O'Reilly, Max de Marzi and Janit Anjaria, for their enthusiastic and detailed feedback.

CHAPTER 1

Introducing Knowledge Graphs

We're overwhelmed by data. It's everywhere and being collected at a fantastic rate and stored at substantial cost. But we're not necessarily getting value from that data, though there is significant value in it, if only we could understand it.

All is not lost. Over the last decade, a new category of technology based on graphs has moved from obscurity to prominence. Graphs have come to underpin everything from consumer-facing systems like navigation and social networks to critical infrastructure like supply chains and power grids.

These important graph use cases have reached a common conclusion: applying knowledge in context is the most powerful tool that most businesses have. A set of patterns and practices called *knowledge graphs* has been emerging to help understand data in context, where the context is represented as a graph of connected data items. With knowledge graphs, there is hope that you can distill business value from data. This book is an attempt to show how it can be done.

Knowledge graphs are useful because they provide contextualized understanding of data. Context derives from the layer of metadata (graph topology and other features) that provides rules for structure and interpretation. This book shows how the connected context provided by knowledge graphs enables you to extract greater value from existing data, drive automation and process optimization, improve predictions, and support an agile response to changing business environments.

We wrote this book for technology professionals who are interested in building and operating knowledge graphs within their businesses. In a way, it's a sequel to or expansion of O'Reilly's *Knowledge Graphs: Data in Context for Responsive Businesses*, which we had the pleasure of writing in 2021. That report was intended for chief information officers (CIOs) and chief data officers (CDOs), helping them to